

Index

Introduction of ChIA-PET
data visualization tools

BASIC Browser demo

WashU EpiGenome
Browser

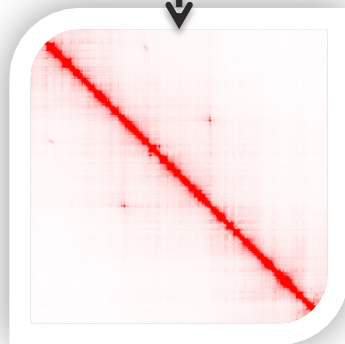
type	local	online	desktop	online	desktop	desktop	desktop
name	BASIC browser	WashU EpiGenome Browser	JuiceBox	HiGlass	IGV	IGB	Sushi (R)
coverage	yes	yes	no	no	yes	yes	yes
peak	yes	yes	no	no	yes	yes	yes
loop	yes	yes	no	no	no	no	yes
heat map	no	yes	yes	yes	no	no	yes

ChIA-PET data visualization tools

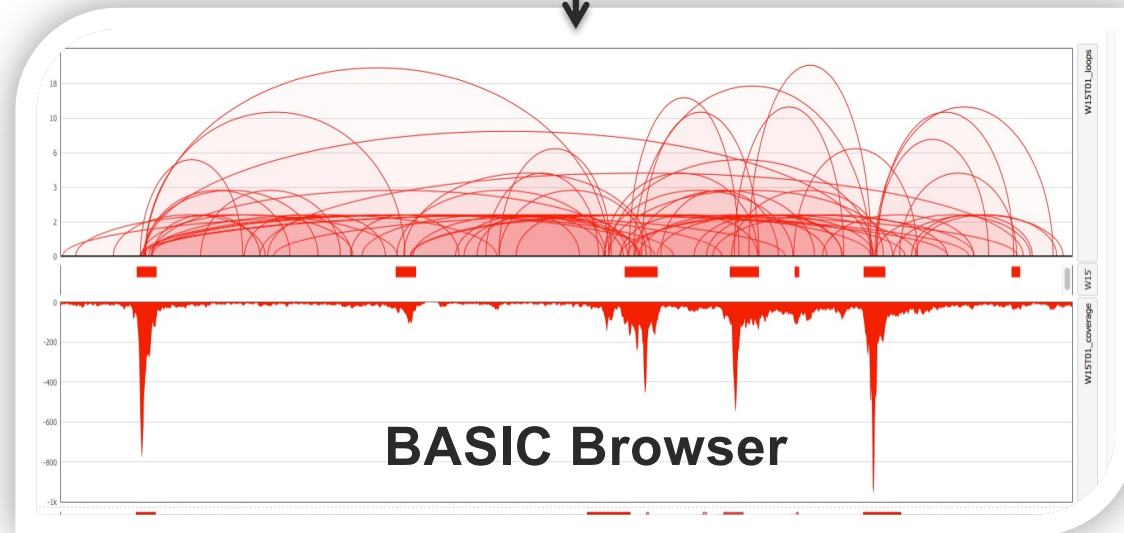
BASIC browser

“Products” of ChIA-PET

ChIA-PET
pipeline



heatmap



loop

peak

coverage

“Products” of ChIA-PET

Loop (interaction)

chr2R	9370601	9370766	chr2R	9390944	9391356	2
chr2L	18152518	18152654	chr3L	9687352	9687497	2
chrX	2546545	2547001	chrX	2555098	2555565	2
chr3R	5188759	5189041	chrX	19191913	19192011	2
chr2L	5986387	5987560	chr2L	5998822	5999526	6
chr2R	5968200	5968352	chr2R	12703724	12703831	2

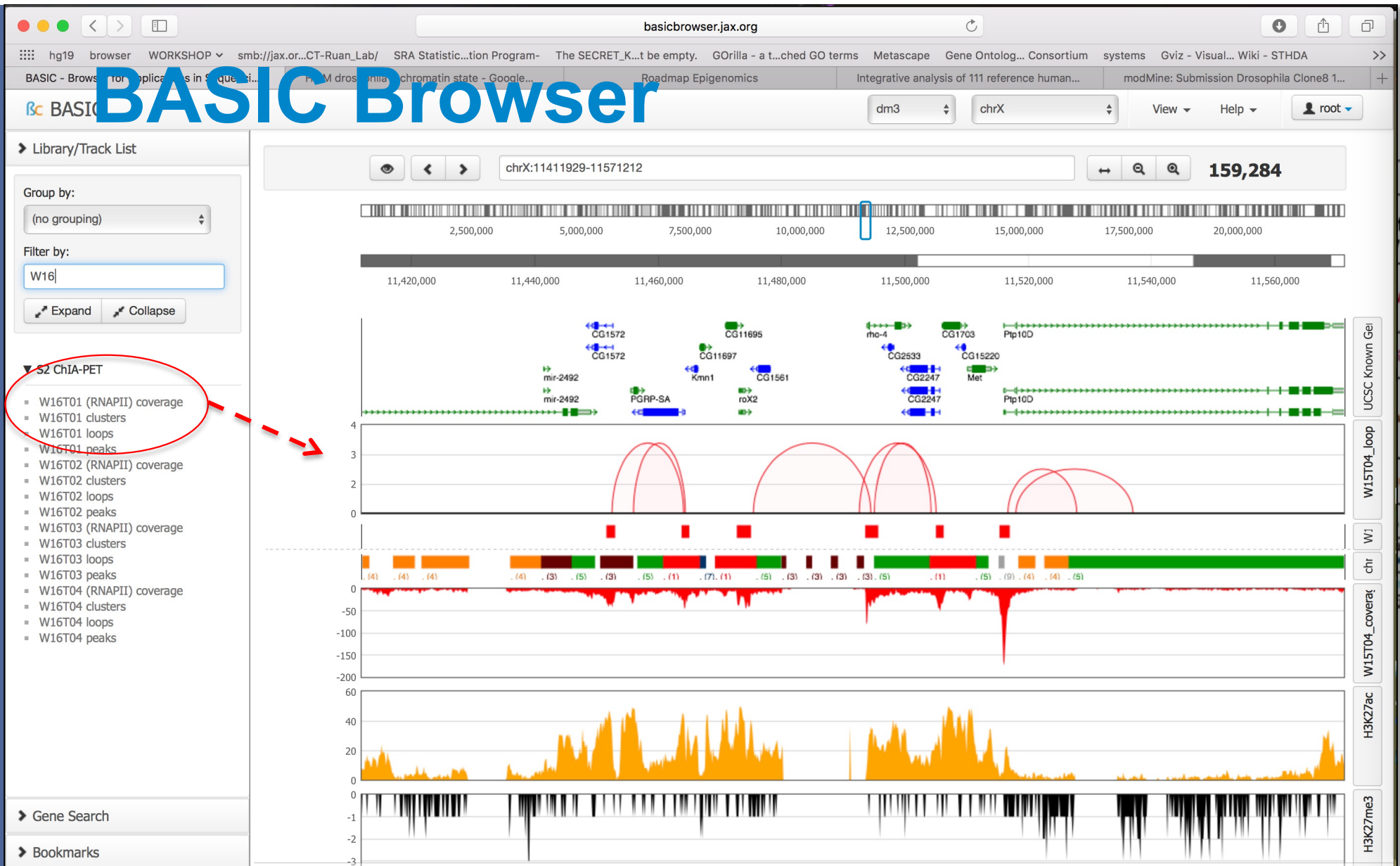
Coverage (bedgraph)

Chr2L	5089	5095	1
Chr2L	5095	5098	4
chr2L	5098	5126	5
chr2L	5126	5141	6
chr2L	5141	5142	7
chr2L	5142	5146	8

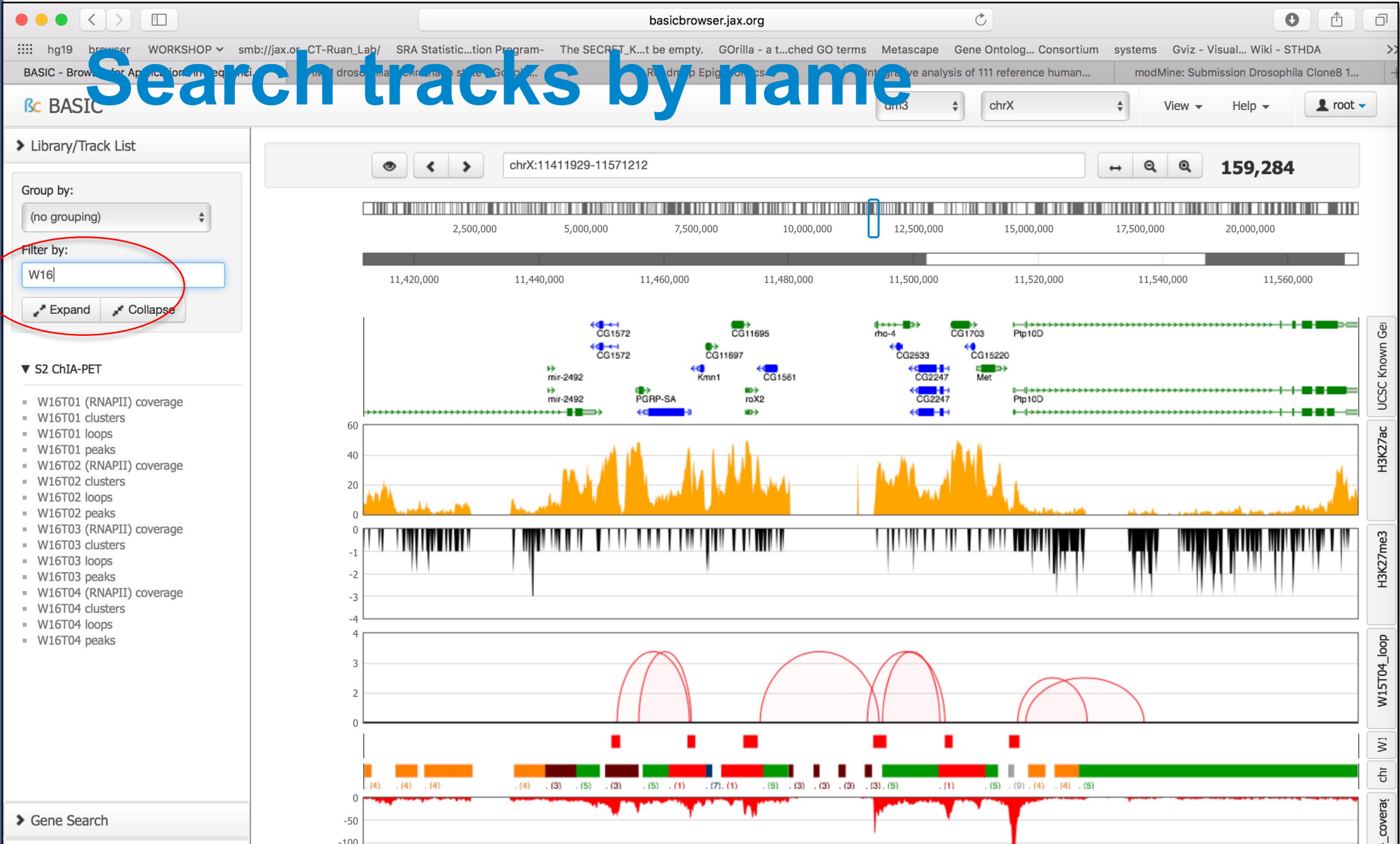
Peak (bed)

chr2L	18285	19151
chr2L	19683	20085
chr2L	66491	67635
chr2L	72120	74214
chr2L	87166	87989

BASIC Browser



Search tracks by name



Search locus by gene name

basicbrowser.jax.org

Library/Track List

Gene Search

roX2 Search

dosage compensat Search

☒ Molecular Function
☒ Biological Process
☒ Cellular Component
☒ Primary
☐ Secondary

chrX:11411929-11571212 159,284

2,500,000 5,000,000 7,500,000 10,000,000 12,500,000 15,000,000 17,500,000 20,000,000

11,420,000 11,440,000 11,460,000 11,480,000 11,500,000 11,520,000 11,540,000 11,560,000

UCSC Known Genes

H3K27ac

H3K27me3

D4_loop

Log Search result for [roX2] x

Symbol	Function	Accession	Location	Strand
roX2	1. X chromosome (C-1)	roX2	chrX:11473129-11474134	+
	2. dosage compensation (P-1)	roX2	chrX:11473121-11474134	+
	3. dosage compensation by hyperactivation of X chromosome (P-1)	roX2	chrX:11473089-11474205	+
	4. X chromosome located dosage compensation complex, transcription activating (C-1)	roX2	chrX:11473098-11474466	+
	5. dosage compensation complex assembly involved in dosage compensation by hyperactivation of X chromosome (P-1)			
	6. negative regulation of chromatin silencing (P-1)			

Search gene by GO function

basicbrowser.jax.org

Library/Track List

Gene Search

by name Search

dosage compensat Search

☒ Molecular Function
☒ Biological Process
☒ Cellular Component

☒ Primary
☐ Secondary

chrX:11411929-11571212 159,284

2,500,000 5,000,000 7,500,000 10,000,000 12,500,000 15,000,000 17,500,000 20,000,000

11,420,000 11,440,000 11,460,000 11,480,000 11,500,000 11,520,000 11,540,000 11,560,000

CG1572 CG1572 CG11695 CG11697 CG1561 roX2 Ptp10D Ptp10D

mir-2492 mir-2492 PGRP-SA Kmn1 roX2

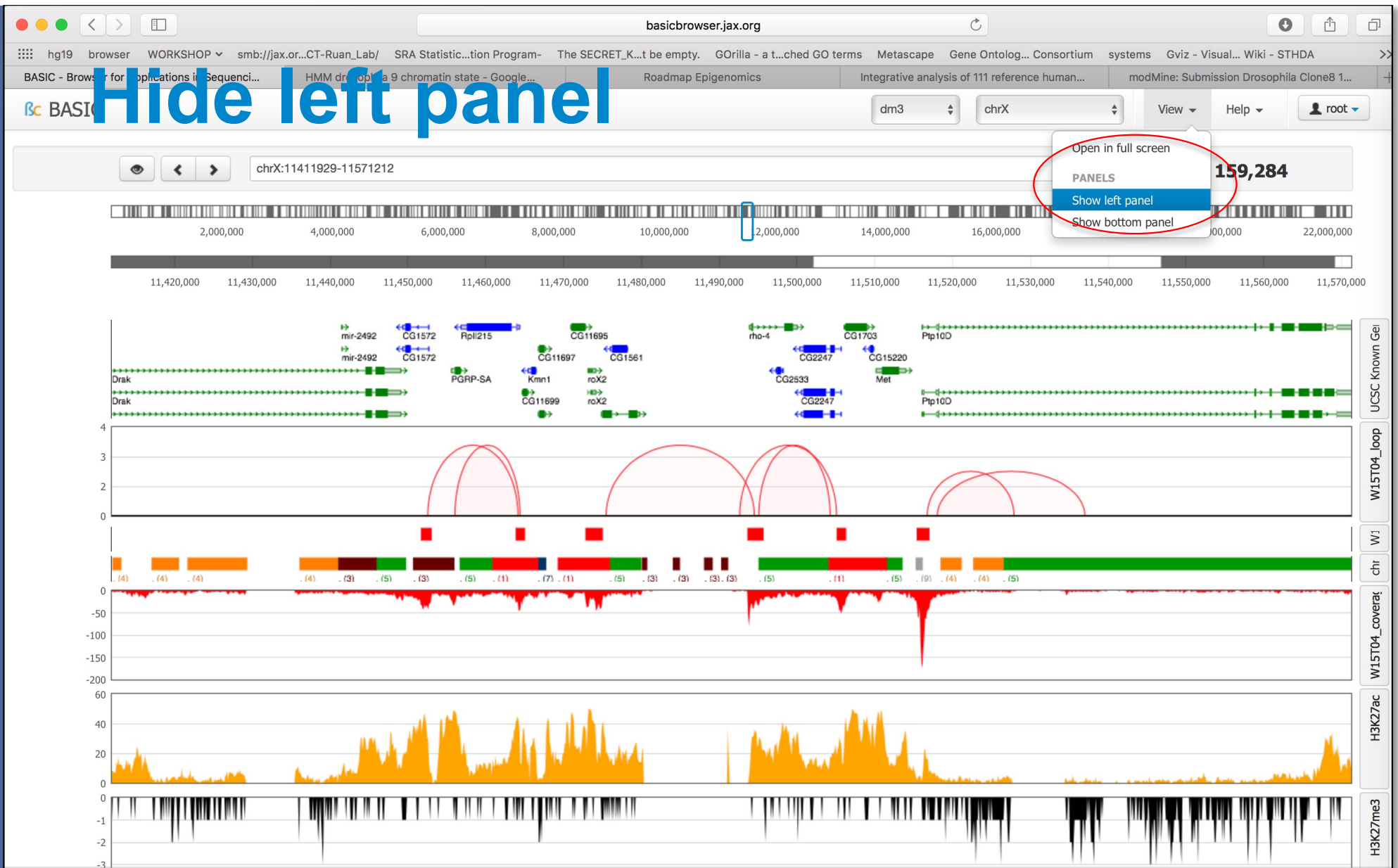
60

Log Search result for [roX2] Search result for [dosage compensation]

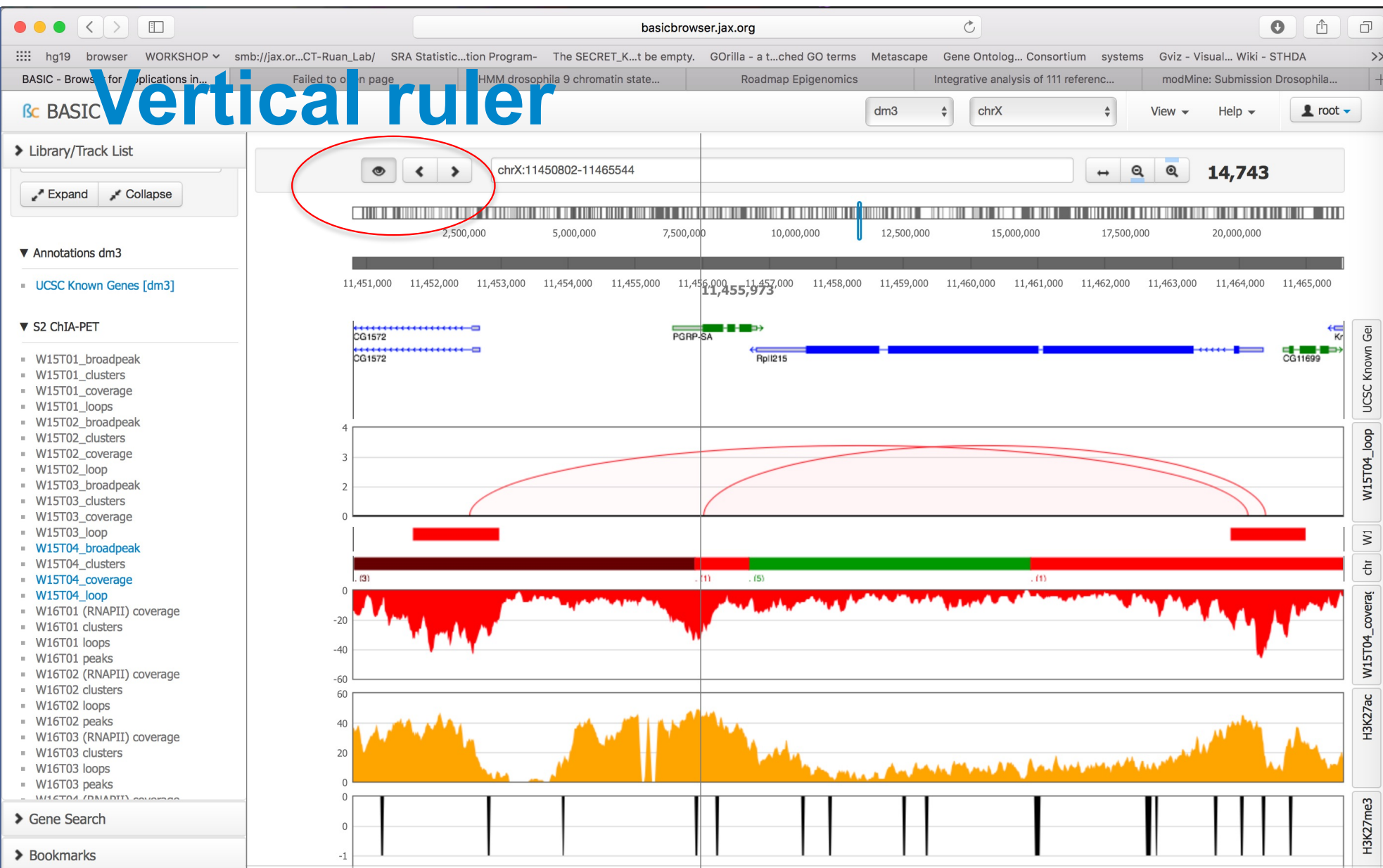
dnp	- negative regulation of transcription from RNA polymerase II promoter (P-1) - RNA polymerase II core promoter proximal region sequence-specific DNA binding transcription factor activity involved in negative regulation of transcription (F-1) - positive regulation of neuroblast proliferation (P-1) - DNA binding (F-1) - sequence-specific DNA binding transcription factor activity (F-1) - nucleus (C-1) - Notch signaling pathway (P-1) - sex determination (P-1) - sex determination, establishment of X:A ratio (P-1) - dosage compensation (P-1) - adult locomotory behavior (P-1) - larval locomotory behavior (P-1) - negative regulation of transcription, DNA-dependent (P-1) - protein dimerization activity (F-1) - dendrite morphogenesis (P-1)	dnp	chr2R:4116476-4119833	-
mle	- chromatin (C-1) - X chromosome (C-1) - nucleic acid binding (F-1) - chromatin binding (F-1) - RNA helicase activity (F-1) - double-stranded RNA binding (F-1)	mle	chr2R:1862243-1868278	-
		mle	chr2R:1862599-1868284	-
		mle	chr2R:1862362-1867607	-

Bookmarks

Hide left panel



Vertical ruler



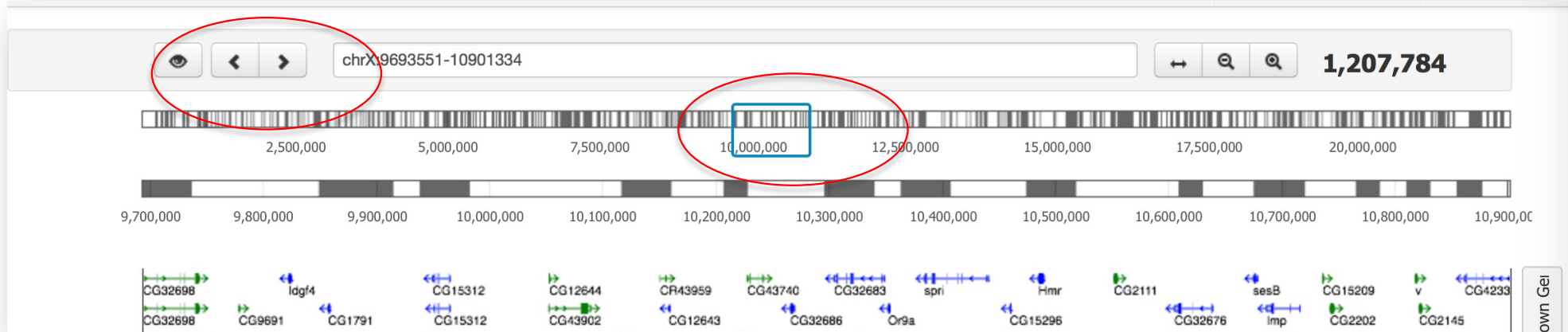
Input gene name or coordinate

The image displays two screenshots of the BASIC browser interface, demonstrating how to search for a gene or coordinate.

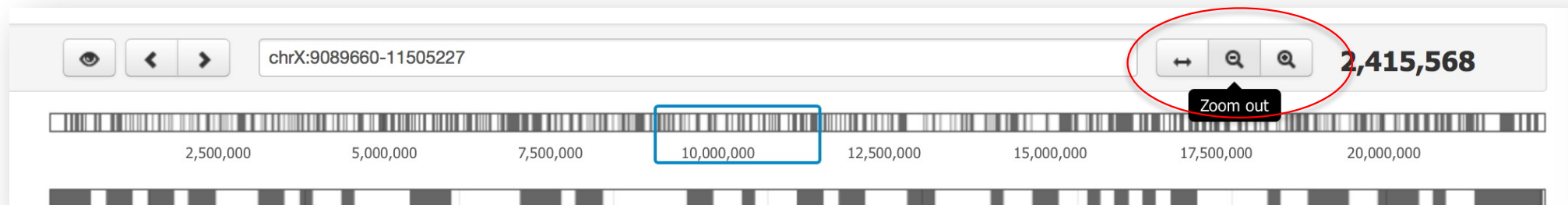
Top Screenshot: The search bar contains the text "roX2". The results show a genomic track with a blue vertical line indicating the position of the gene. The track is labeled "roX2" and "Rpl215". The coordinates range from 11,451,000 to 11,465,000. The track is labeled "roX2" and "Rpl215". The track is labeled "roX2" and "Rpl215".

Bottom Screenshot: The search bar contains the text "chrX:11473089-11474466". The results show a genomic track with a blue vertical line indicating the position of the gene. The track is labeled "roX2" and "roX2". The coordinates range from 11,473,100 to 11,474,400. The track is labeled "roX2" and "roX2".

Select region by select and shift slide-window



Zoom in/out and entire chromosome



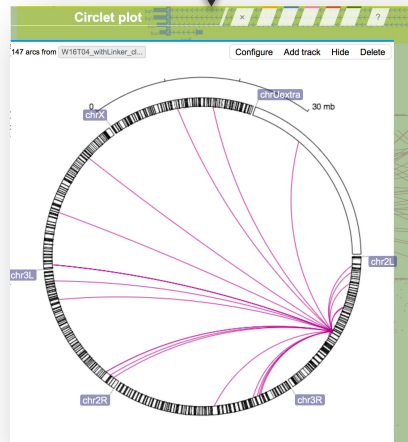
WashU
EpiGenome
Browser

<http://epigenomegateway.wustl.edu>

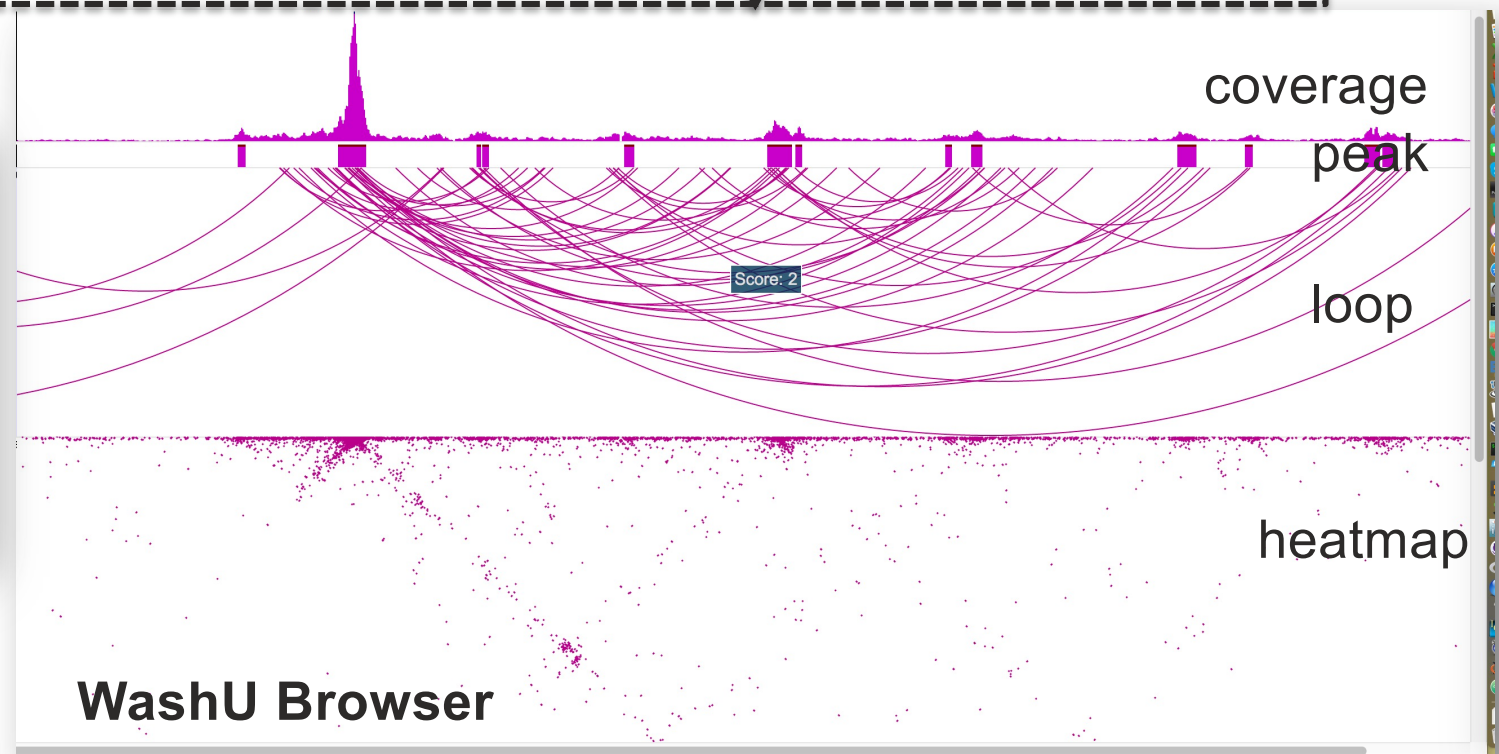
WashU EpiGenome Browser

“Products” of ChIA-PET

ChIA-PET
pipeline



Circlet plot



Data preparation for WashU Browser

Coverage (bedgraph)

Chr2L	5089	5095	1
Chr2L	5095	5098	4
chr2L	5098	5126	5
chr2L	5126	5141	6
chr2L	5141	5142	7
chr2L	5142	5146	8

Interaction (loop)

chr2R:9370601-9370766	chr2R:9390944-9391356	2
chr2L:18152518-18152654	chr3L:9687352-9687497	2
chrX:2546545-2547001	chrX:2555098-2555565	2
chr3R:5188759-5189041	chrX:19191913-19192011	2
chr2L:5986387-5987560	chr2L:5998822-5999526	6

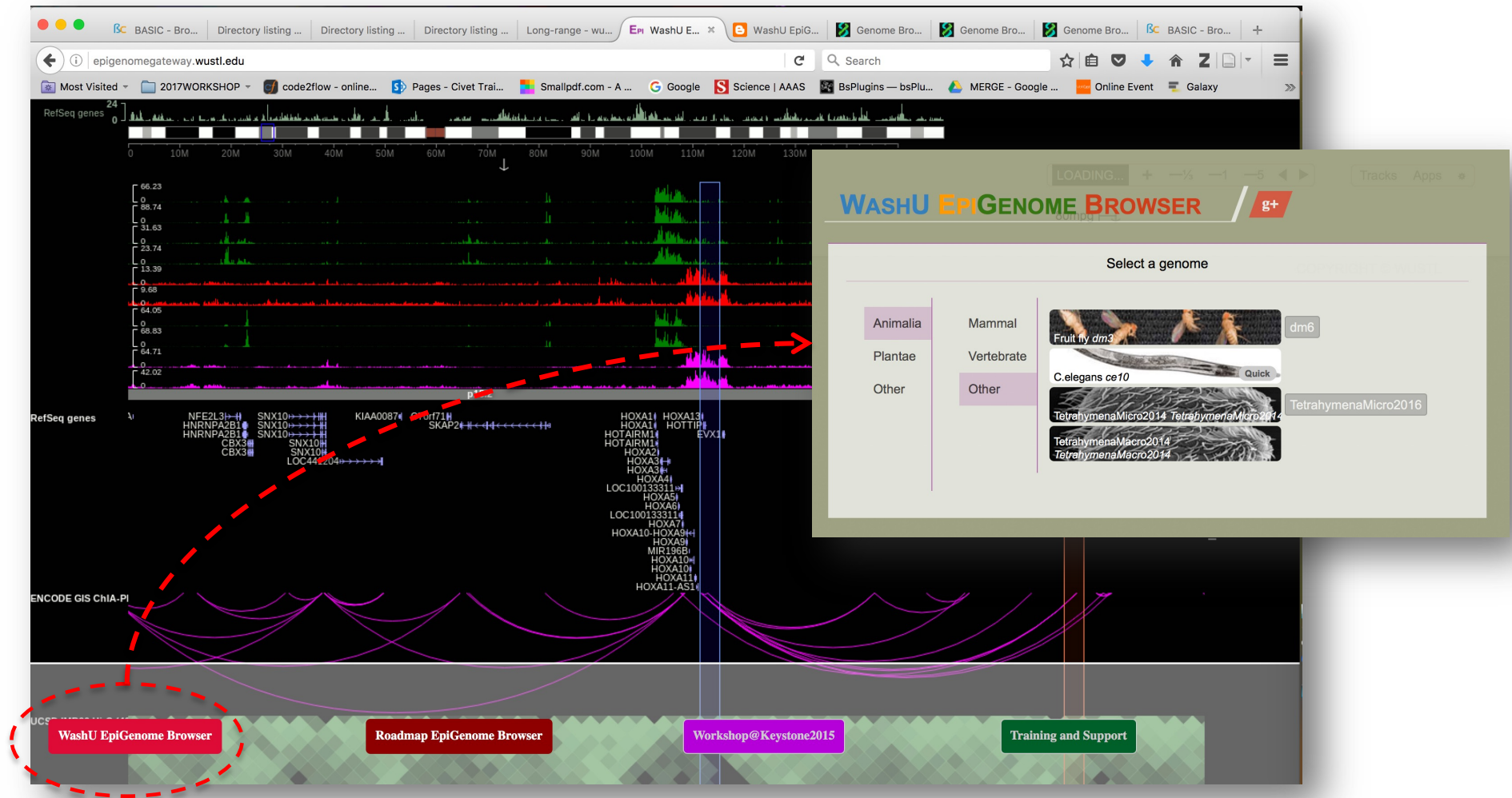
Raw contact (heatmap)

chr2L,21757601,21757697	chr2R,20299479,20299587	1
chr3R,10195326,10195444	chr3R,10195813,10195878	1
chr3R,16377350,16377422	chr3R,16378032,16378139	1
chr3R,25609345,25609479	chr3R,25609345,25609479	1
chr2L,10399320,10399426	chr3L,1327482,1327632	1
chrX,21190115,21190232	chrX,21190406,21190557	1
chr3L,8449665,8449720	chr3L,8449666,8449736	1

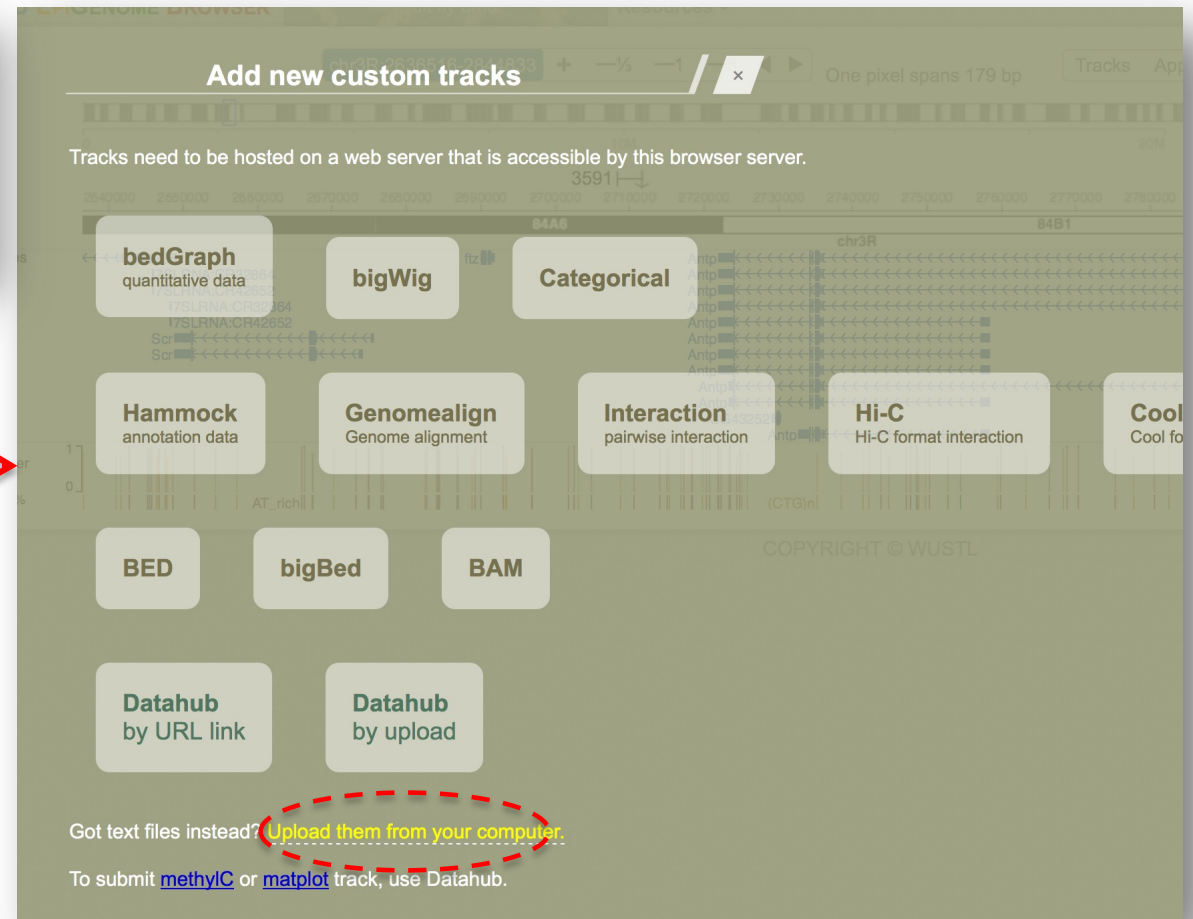
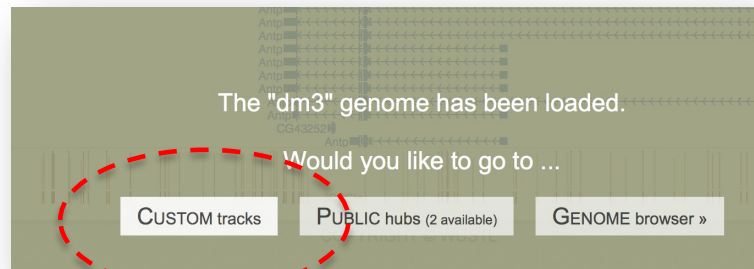
Peak (bed)

chr2L	18285	19151
chr2L	19683	20085
chr2L	66491	67635
chr2L	72120	74214
chr2L	87166	87989

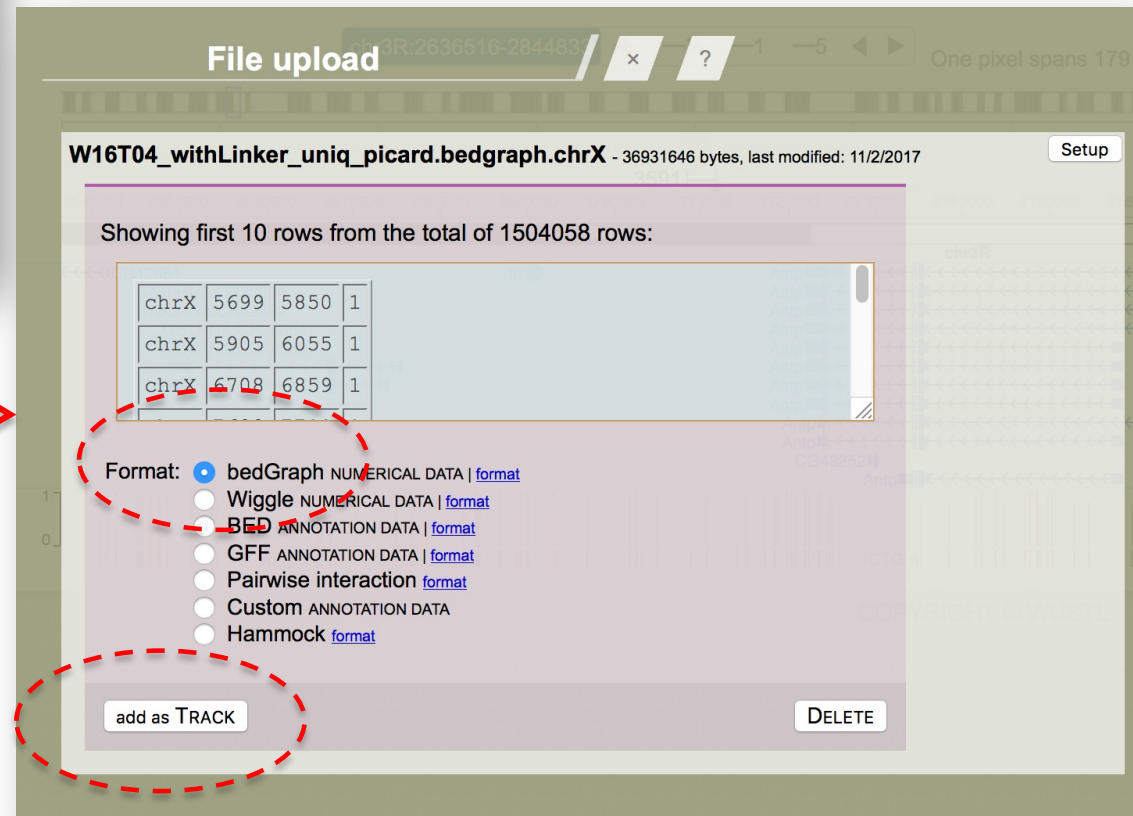
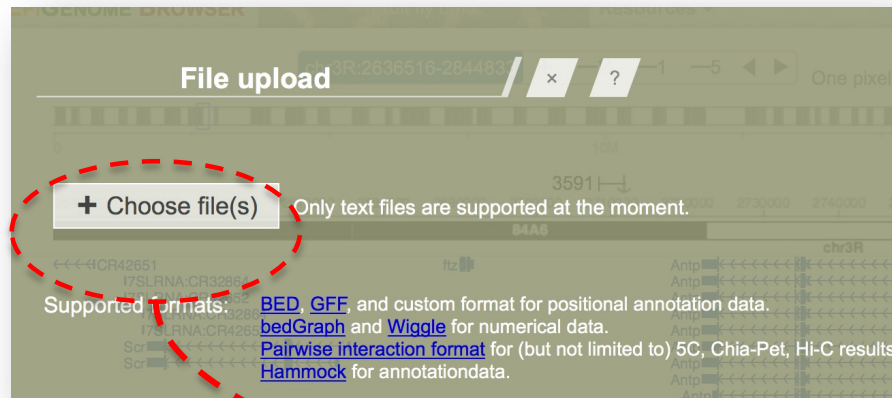
Access WashU EpiGenome Browser



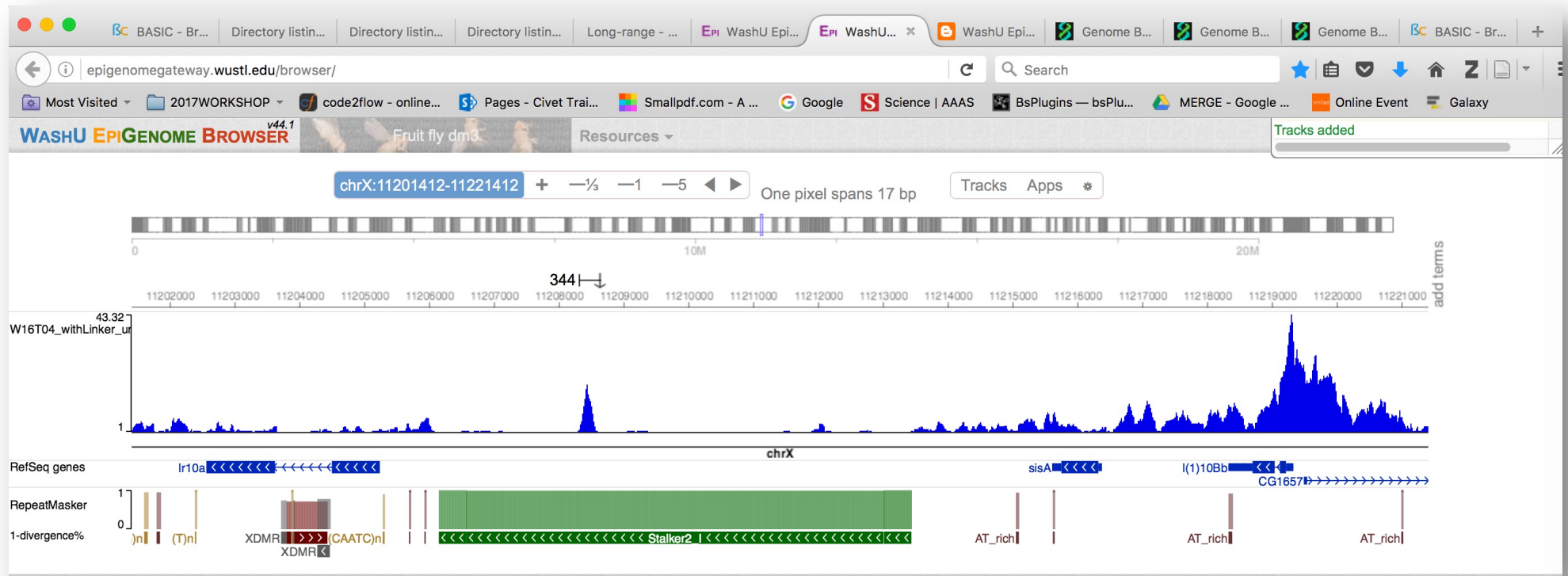
Upload a coverage (bedgraph)



Upload a coverage (bedgraph)



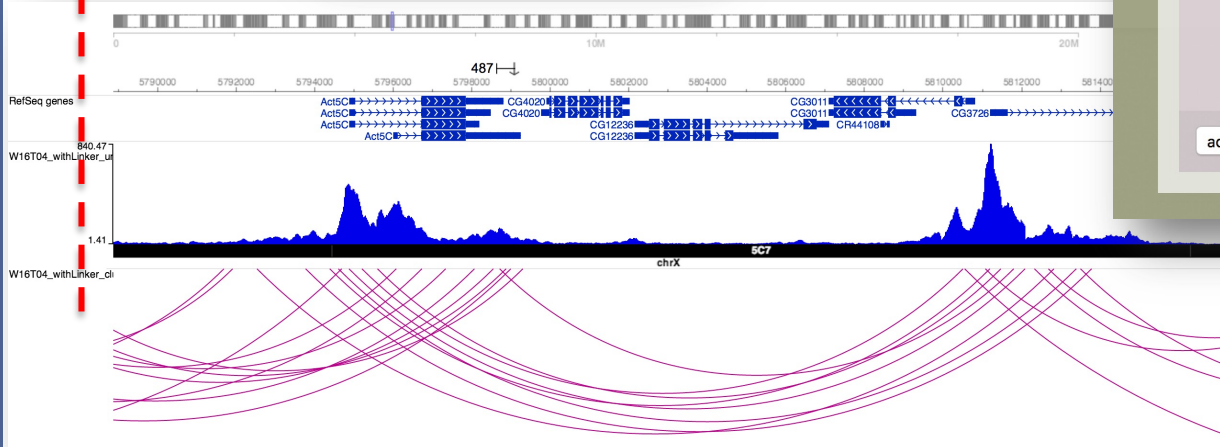
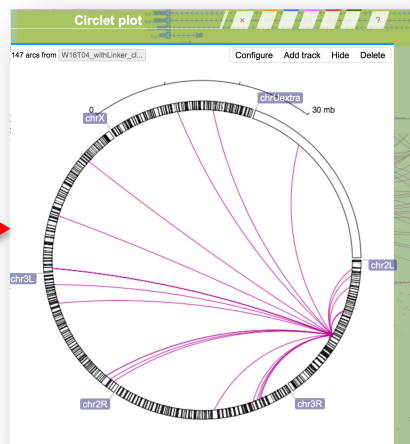
Upload a coverage (bedgraph)



Upload a interaction (loop)

The screenshot displays the WashU EpiGenome Browser interface. The main track shows a genomic region on chromosome X (chrX:11201412-11221412) with a scale of 10M. Below the main track, several tracks are visible: W16T04_withLinker_ur, RefSeq genes (including Ir10a and Stalker2), RepeatMasker, and 1-divergence%. A red dashed circle highlights the 'Tracks' button in the top right corner. A dropdown menu is open, showing '1 TOTAL / 1 SHOWN' and a 'CLICK FOR TRACK TABLE' link. The menu also includes a search bar for 'track name' and a 'Find ?' button. Below the search bar, the menu lists 'PUBLIC track hubs', 'ANNOTATION tracks', and 'CUSTOM tracks (1)'. A red dashed arrow points from the 'CUSTOM tracks (1)' option to a separate window that displays '1 TOTAL' and a 'List of all' button, with a '+ Add new tracks' button at the bottom.

Upload a interaction (loop)



File upload

W16T04_withLinker_clusters.washU - 690523 bytes, last modified: 11/2/2017

Showing first 10 rows from the total of 14536 rows:

chr2R:9370601-9370766	chr2R:9390944-9391356	2
chr2L:18152518-18152654	chr3L:9687352-9687497	2
chrX:2546545-2547001	chrX:2555098-2555565	2

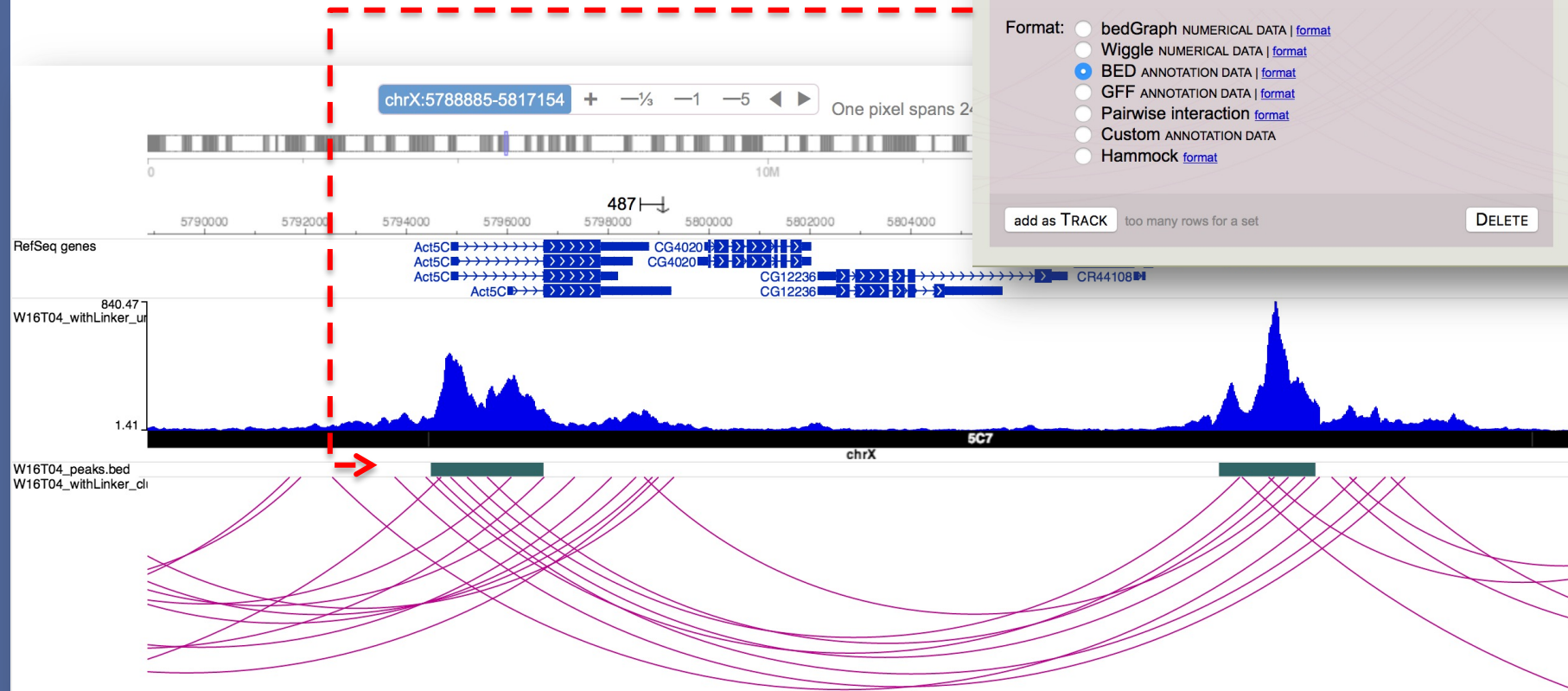
Format:

- ☐ bedGraph NUMERICAL DATA | [format](#)
- ☐ Wiggle NUMERICAL DATA | [format](#)
- ☐ BED ANNOTATION DATA | [format](#)
- ☐ GFF ANNOTATION DATA | [format](#)
- ☒ Pairwise interaction [format](#)
- ☐ Custom ANNOTATION DATA
- ☐ Hammock [format](#)

add as TRACK

DELETE

Peak track upload



W16T04_peaks.bed

- 245853 bytes, last modified: 11/2/2017

Setup

Showing first 10 rows from the total of 5554 rows:

chr2L	18285	19151	NA_peak_1	14.71775
chr2L	19683	20085	NA_peak_2	8.48001
chr2L	66491	67635	NA_peak_3	38.48325

Format:

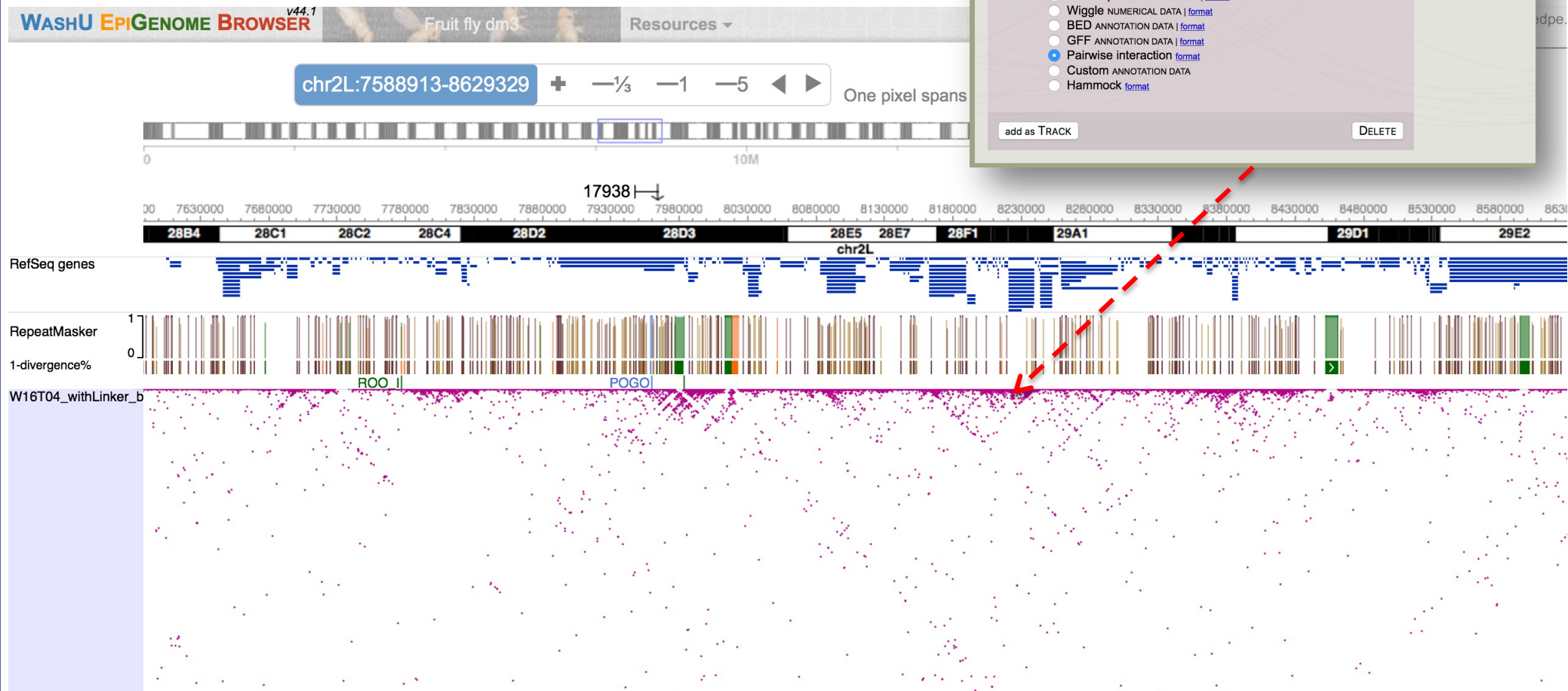
☐ bedGraph NUMERICAL DATA | [format](#)
☐ Wiggle NUMERICAL DATA | [format](#)
☒ BED ANNOTATION DATA | [format](#)
☐ GFF ANNOTATION DATA | [format](#)
☐ Pairwise interaction [format](#)
☐ Custom ANNOTATION DATA
 ☐ Hammock [format](#)

add as TRACK

too many rows for a set

DELETE

Heatmap track upload





National Human Genome
Research Institute

References and acknowledgment

- WashU EpiGenome Browser:
<http://epigenomegateway.wustl.edu>